DAY 04 OF 16 · 3D & 6DOF STREAMING

Point clouds, formats & bytes

The same 190,000 points can weigh 6 MB or 1 MB. Today we learn why, because bytes are what we stream.

Anatomy of a point | ASCII vs binary | PLY, LAS, NPY | Quantisation | Bytes → bandwidth

Book: Ch. 3–4 · Code: day04_point_clouds_formats/

ANATOMY OF A COLOURED POINT

How many bytes is one point?

```
3 × float32 (12 B) + 3 × uint8 (3 B) = 15 bytes
```

x, y, z

4 bytes each as 32-bit floats: about 7 significant digits.

r, g, b

1 byte each, values 0–255.

extras

Intensity, class, normal, time: each adds bytes per point.

LAB RESULT (190K POINTS)

Same points, very different sizes

Format	Size	Bytes / point	Load time
ASCII .txt (x y z r g b c)	5.9 MB	32.6	≈ 80 ms
PLY ascii	6.0 MB	33.5	≈ 900 ms
PLY binary (float32 + uint8)	2.7 MB	15.0	≈ 30 ms
NumPy .npy (7 × float32)	4.3 MB	24.0	≈ 1 ms
Quantised 1 mm + zlib	1.0 MB	5.7	≈ 8 ms

LAS (the LiDAR standard) stores integers with a scale and offset, which is already a form of quantisation. Install laspy to add it.

CHOOSING A FORMAT

Each format answers a different need

PLY

Simple, flexible attributes, binary or ASCII. Our default.

01

LAS / LAZ

LiDAR standard: classes, GPS time, intensity. LAZ is losslessly compressed.

02

E57

Scanner exchange format with images and poses.

03

Streaming formats

Potree octree, Cesium 3D Tiles, MPEG G-PCC / V-PCC: built for progressive delivery.

04

BYTES → BANDWIDTH

Why raw 3D does not fit through a network

```
points × bytes × fps × 8 = bits per second
```

100k pts

× 15 B × 30 fps ≈ 360 Mbit/s.

1M pts

× 15 B × 30 fps ≈ 3.6 Gbit/s.

Compare

A 4K movie stream is about 15 Mbit/s.

HANDS-ON LAB · COPY, RUN, LOOK

Measure it yourself

RUN IT

```
cd day04_point_clouds_formats
python formats_and_sizes.py
pip install laspy   # optional
python formats_and_sizes.py
```

→ Read the size table in the terminal.

→ Change the quantisation step to 1 cm.

→ Compute points per frame for a 25 Mbit/s link.

Three things to remember

1. A coloured point costs about 15 bytes in binary; text formats cost twice as much or more.
2. Quantising coordinates (snapping to a grid) shrinks data with little visible loss.
3. points × bytes × fps × 8 = bitrate. Raw volumetric video needs hundreds of Mbit/s.

BEFORE NEXT SESSION

Finish Lab L2. Next: points are not the only way to represent 3D.